TSYP14 TECHNICAL CHALLENGE · TRACK 1 — SECURE GROUND-LINK TESTBED · ANONYMOUS SUBMISSION

SENTINEL-LINK: authenticating stored satellite downlinks — cryptography where the link allows it, physics where it does not

Team code: assigned by the administrative committee. All numbers in this report are produced by `py -m sentinel bench` on 17 held-out SatNOGS observations (results/benchmark.json, generated 2026-10-23T14:07:52+00:00).

99.3%
replayed / injected frames detected (581/585), decoded from attacked real audio

0.71%
frame false-alarm rate (attacked + clean passes)

90% (n=20)
jammers detected at J/N ≥ 0 dB (all kinds, hidden parameters)

7.06 µs
SDLS verify per frame, 22 B overhead, 100% replay rejection

Summary. Open ground networks record satellite downlinks that carry no authentication. We built a ground-station testbed that (i) *prevents* replay and injection on a link we control with CCSDS Space Data Link Security, validated byte-for-byte against NASA CryptoLib; (ii) *authenticates stored recordings* with a station-signed per-second Merkle manifest; and (iii) *detects* the three attacks of the standard test on legacy, unauthenticated downlinks using evidence an attacker cannot easily fake: the spacecraft's own frame counter against reception time, the Doppler and level continuity of the received carrier, the consistency of housekeeping telemetry, and calibrated spectral evidence of added interference. Everything runs on a laptop CPU, on authentic SatNOGS recordings of AO-73 (FUNcube-1) that we demodulate and FEC-decode ourselves.

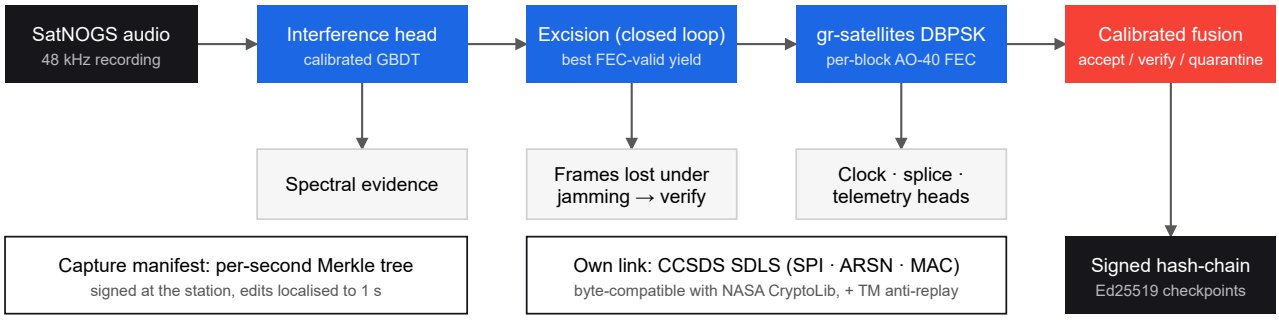
1 · Threat model and segment

Segment: RF downlink → ground receiver → **recording archive** → decoder → product pipeline. Adversary capabilities considered: passive recorder (SatNOGS publishes recordings), replay/splice attacker, signal-level attacker adding narrowband interference, and an informed attacker who rewrites counters and re-encodes FEC. Radiated attacks are out of scope (file-based testbed, per the safety rule).

STRIDE	ON THIS SEGMENT	CONTROL
Spoofing	forged frame, plausible content	SDLS MAC (prevent) · telemetry & clock heads (detect)
Tampering	edited stored recording, altered counter	signed Merkle manifest (localised) · clock head
Repudiation	disputed decisions	hash-chained ledger with Ed25519 checkpoints
Info. disclosure	cleartext telemetry	SDLS AES-256-GCM on own link
Denial of service	narrowband jammer	interference head · closed-loop excision · lost-frame accounting
Elevation	poisoning anti-replay state	window advanced only after MAC verification

2 · Architecture and trust anchor

NASA OnAIR's Agent and VehicleRepresentation run their telemetry feasibility tests on decoded AO-73 frames. Limits come from training passes; our OnAIR plugin checks the spacecraft frame counter against the clock. These two checks expose different evidence: plausible replayed telemetry can pass feasibility limits while the unchanged counter fails the timing check. The web console exposes the per-frame outputs alongside the other heads.



Software trust anchor. One 256-bit master secret; per-SA keys by HKDF-SHA-256 (interface identical to an HSM's derive call); Ed25519 station key. **SDLS:** TM transfer frame | SPI | ARSN | data | 128-bit MAC | FECF. Our frames are **byte-identical to NASA CryptoLib's** (v1.7.5, the version shipped with NOS3) and each library accepts the other's frames and rejects forgeries (`tests/test_cryptolib_interop.py` , real AO-73 frames).

> **Finding.** In that CryptoLib version the anti-replay check is applied on TC (uplink) only: the TM and AOS processing paths accept a replayed frame whose MAC verifies. Our receiver adds an ARSN sliding window (W = 64, bitmap for late frames, window advanced only after MAC success): replayed TM frames are rejected. On 2,085 protected real frames: replay rejection 100.0%, bit-flip rejection 100.0%, overhead 22 B (8.59%) and 7.06 µs per frame (AES-GCM: 34 B, 4.12 µs).

Stored-signal authentication. At capture the station hashes each second of 16-bit PCM, builds a Merkle tree and signs root + capture context. Clean recordings verify; every attacked recording fails and the edited seconds are listed (e.g. 86 s flagged for a replay at [312.1, 347.0] s plus a jammer); signing a 6-min recording takes ≈0.26 s and the manifest is ≈47.9 kB.

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3 · Signal processing

SatNOGS audio (48 kHz) of AO-73's 1200 Bd DBPSK downlink. The carrier offset is estimated from the two BPSK lobes at f_c±600 Hz. gr-satellites demodulates (FLL, clock recovery, differential detection); we dump its symbol stream, search the 65-bit distributed AO-40 syncword, and decode *each 5,200-symbol block separately* with gr-satellites' AO-40 deframer (Viterbi, descrambler, two RS codes). This binds every frame to its exact start time — gr-satellites' own timestamps are wall-clock — and recovers **1,226 frames from the 30 recordings vs 1,143 decoded live by the stations**. The same pass yields the carrier (residual Doppler) and level tracks at 10 Hz. Spectral evidence: 4,096-point STFT, prominence over a running baseline, spectral kurtosis, persistence, BPSK-line balance, plus 30 s context features for frequency-static emitters and narrowband plateaus. Pass geometry is propagated with SGP4 from each observation's TLE (spot check: max elevation 44.5° vs 45° in the SatNOGS metadata, rise azimuth 183° vs 183°).

4 · Detection

Clock head. AO-73 frames carry a 6-bit type (0–23) and a 24-bit counter that increments once per 24-frame set, so P = 24·seq + type advances once per frame. A robust linear model fitted on training passes gives an on-board frame period of **4.9885 s**; leave-one-pass-out, every authentic frame of an unseen pass lies within one frame of it (Fig. 1). Replayed frames lag by the replay delay; counter evidence is split into "behind" (replay) and "ahead".

Splice head. Steps of the carrier and level tracks at frame boundaries and the frame's offset from a robust fit of the whole-pass Doppler curve. **Telemetry head.** Neighbour-relative deviations of 12 housekeeping channels scored with PyOD ECOD. **Interference head.** Calibrated gradient boosting on spectral evidence, with a per-station site survey masking recurring emitters learned from the station's past recordings.

Fusion. Logistic regression on log-scaled evidence (no class re-weighting, so outputs are calibrated: ECE 0.0231); ACCEPT < 0.5 ≤ VERIFY < 0.9 ≤ QUARANTINE.

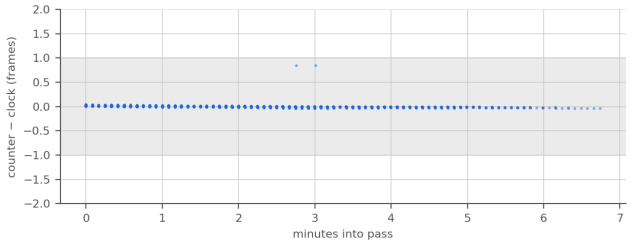


Fig. 1 — Counter minus clock model for every authentic frame of all 30 passes, model fitted without the pass shown (leave-one-pass-out).

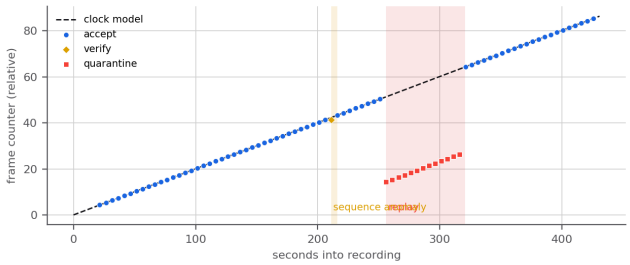


Fig. 2 — Held-out pass 15123656 after the standard attack, re-decoded: 14/14 attacked frames flagged, 0 false alarms in 82 frames.

5 · Adversarial validation and response

Protocol: models trained on 13 early observations; evaluated on 17 later observations / unseen stations. For each, three random draws of the standard test (replay of 5–20 frames, one sequence anomaly — duplicated or swapped slot — and a CW / narrowband-noise / swept / pulsed jammer at random frequency and hidden J/S) plus one replay spliced from a different pass, all applied to the real audio and **decoded again by the full chain**.

METRIC	VALUE
Attacked frames detected	99.3% (581/585)
Events: standard replay · standard anomaly · cross-pass replay	44/44 · 41/41 · 16/16
False-alarm rate: attacked passes · clean passes	0.85% · 0.29%
ROC-AUC (frames) · ECE	0.9996 · 0.0231
Evasion: counter forged +1 · counter forged far · counter-consistent forged content	88.2% · 100.0% · 64.7%
Jammed recordings detected · clean recordings alarmed	77.1% (n=48) · 7/17
Decision latency per frame after decoding · peak memory	2.414 ms · 43.8 MB

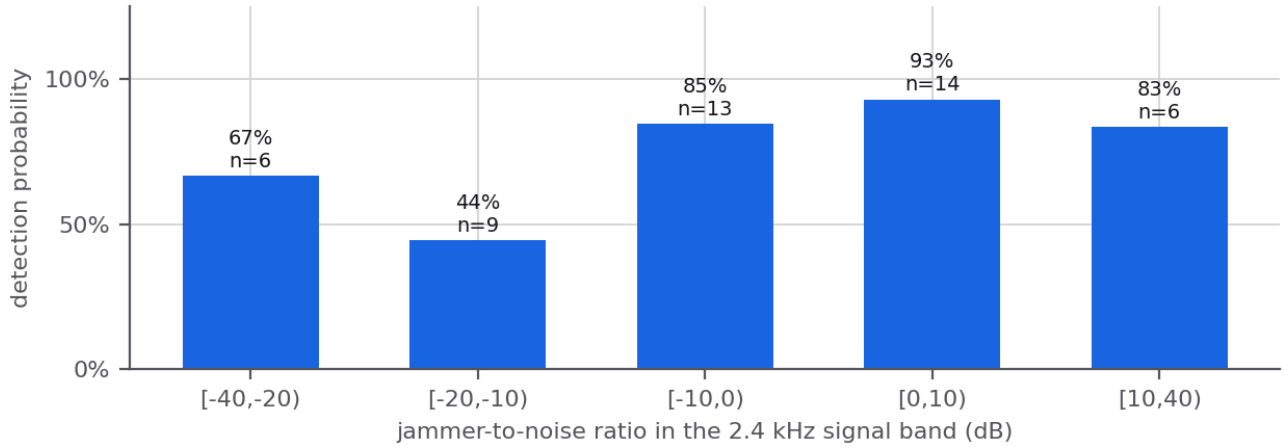


Fig. 3 — Interference detection vs jammer-to-noise ratio (held-out recordings, all jammer kinds).

Response. Frames are accepted, sent for verification, or quarantined and logged. Detected interference triggers a closed-loop excision: band-stop, time–frequency mask and no action are each decoded and the response yielding the most Reed–Solomon-valid frames is kept (reproducible example, swept tone, J/S +6 dB, 900 Hz below the carrier on obs 15254375: frames decoded per response {'none': 0, 'band_stop': 0, 'tf_mask': 0}, chosen *none*; clean reference 60). Frame positions that the clock says were transmitted but are missing inside an interference alarm are reported as unaccounted (a jammer can hide an injected or dropped frame). In 8 benchmark cases the excision kept every decodable frame.

Limits we measured. Jammers well below the noise in the signal band are often missed; at these levels they also do not degrade decoding. A jammer placed inside a station's known-RFI range and below that emitter's level evades the site survey. A frame forged with a clock-consistent counter and telemetry close to its neighbours is not detectable without cryptography — the argument for SDLS on links we control and signed capture at the station.

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6 · Official reference data and reproducibility

30 SatNOGS Network observations of AO-73 (5 stations, 29 Sep – 22 Oct 2026; audio, station decodes, TLE, station position), every file SHA-256-recorded. Note: the Network API filters by `norad_cat_id`; the `satellite__norad_cat_id` parameter is silently ignored. One command per step: `py -m sentinel fetch | decode | train | bench`; held-out analysis: `py -m sentinel analyse heldout.wav` (start time optional — inferred from the on-board clock). A web console (FastAPI + Next.js/three.js) shows each pass on a 3D globe from its TLE, runs the attack lab on real audio, analyses uploaded recordings blind, and demonstrates SDLS and the ledger live.

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7 · Feasibility

Laptop CPU only (12 logical cores, no GPU); GNU Radio in WSL2 on Windows or native Linux. Decoding runs at ≈35–70× real time; detection after decoding takes 2.414 ms per frame; the interference head ≈13835.5 ms per 6-min recording; benchmark peak memory 43.8 MB. No purchase is required; an SDR is optional for Phase 2 live capture.

References: CCSDS 355.0-B-2 Space Data Link Security Protocol; NASA CryptoLib (github.com/nasa/CryptoLib); SatNOGS Network (network.satnogs.org); gr-satellites (D. Estévez); AO-40 FEC (P. Karn); PyOD ECOD (Li et al., 2022); C2PA technical specification; SGP4 (Vallado et al.).